

Advanced (Habanero)

Q1. Factor the trinomial $x^2 + 5x + 6$.

- (A) $(x + 2)(x + 3)$
- (B) $(x + 1)(x + 6)$
- (C) $(x - 2)(x - 3)$
- (D) $(x - 1)(x - 6)$
- (E) $(x + 3)(x - 2)$

Q2. Factor the trinomial $x^2 - 7x + 12$.

- (A) $(x - 3)(x - 4)$
- (B) $(x - 1)(x - 6)$
- (C) $(x + 3)(x + 4)$
- (D) $(x + 1)(x + 6)$
- (E) $(x - 2)(x - 5)$

Q3. Factor the trinomial $x^2 + 2x - 15$.

- (A) $(x + 5)(x - 3)$
- (B) $(x + 1)(x - 15)$
- (C) $(x - 5)(x + 3)$
- (D) $(x - 1)(x + 15)$
- (E) $(x + 3)(x + 5)$

Q4. Factor the trinomial $x^2 - 4x - 21$.

- (A) $(x - 7)(x + 3)$
- (B) $(x - 1)(x - 21)$
- (C) $(x + 7)(x - 3)$
- (D) $(x + 1)(x + 21)$
- (E) $(x - 3)(x - 7)$

Q5. Factor the trinomial $x^2 + 10x + 24$.

- (A) $(x + 4)(x + 6)$
- (B) $(x + 1)(x + 24)$
- (C) $(x - 4)(x - 6)$
- (D) $(x - 1)(x - 24)$
- (E) $(x + 6)(x - 4)$

Q6. Factor the trinomial $x^2 - 2x - 15$.

- (A) $(x - 5)(x + 3)$
- (B) $(x - 1)(x - 15)$
- (C) $(x + 5)(x - 3)$
- (D) $(x + 1)(x + 15)$
- (E) $(x - 3)(x + 5)$

Q7. Factor the trinomial $x^2 + 7x + 12$.

- (A) $(x + 3)(x + 4)$
- (B) $(x + 1)(x + 12)$
- (C) $(x - 3)(x - 4)$
- (D) $(x - 1)(x - 12)$
- (E) $(x + 4)(x - 3)$

Q8. Factor the trinomial $x^2 - 9x + 20$.

- (A) $(x - 4)(x - 5)$
- (B) $(x - 1)(x - 20)$
- (C) $(x + 4)(x + 5)$
- (D) $(x + 1)(x + 20)$
- (E) $(x - 5)(x - 4)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $2x^2 + 5x - 3$. Show all steps and provide the factored form.

Q10. Factor the trinomial $3x^2 - 2x - 1$. Show all steps and provide the factored form.

Chef's Tips

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- Emphasize the importance of checking the factored form by multiplying it out.
- Encourage students to use different factoring methods, such as grouping or the quadratic formula.
- Remind students to always write the factored form in the simplest way possible.